

MAYANK AGRAWAL

Full-Stack Engineer | Generative AI, RAG & Agentic Applications

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PROFESSIONAL SUMMARY

Full-Stack Engineer with 3+ years of experience at EY building scalable enterprise applications using React.js, Typescript, Redux Toolkit, RTK Query, ASP.NET, and REST APIs. Experienced in reusable frontend architecture, application performance optimization, automated testing, CI/CD and enterprise workflow development across a large financial platform containing 40+ modules.

Hands-on experience designing Generative AI applications using Python, FastAPI, Langchain, LangGraph, LLM APIs, PostgreSQL, pgvector, hybrid retrieval, reranking, evaluation, guardrails, and agentic workflows. Building end-to-end AI applications that combine enterprise-grade frontend engineering with retrieval pipelines, vector search, API development, persistent memory, observability, and cloud-ready deployment patterns.

CORE SKILLS

Programming: JavaScript (ES6+), Python, TypeScript, C#.

Frontend: React.js, Redux Toolkit, RTK Query, HTML5/CSS3, Formik, Component-Driven Architecture, Performance Optimization (code-splitting, Lazy Loading, memoization), Responsive Design, State Management, Rspack, Vite.

Backend: Node.js, Express.js, FastAPI, ASP.NET Core, RESTful API Design, JWT Authentication, Git, CI/CD (Azure Pipelines, GitHub Actions)

AI/GenAI: Large Language Models, LangChain, LangGraph, Prompt Engineering, Function Calling, RAG (vector databases), Agentic Workflows (Multi-Agent Systems, MCP), Vector Embeddings, Semantic Search, LangSmith, AI Evaluation/Guardrails

Testing & Tools: Jest, React Testing Library, Unit & Integration Testing, E2E Testing, Logging & Monitoring, Agile/Scrum methodologies

PROFESSIONAL EXPERIENCE

Ernst & Young (EY) | Software Engineer (Full-Stack & AI Application Engineering) | July 2023 – Present (Pune, India)

- Engineered reusable component frameworks including dynamic tables, configurable forms, filtering systems, and workflow-driven UI experiences adopted across multiple business modules.
- Optimized frontend performance through lazy loading, memoization and efficient state management, reducing load time by ~25% for data-intensive workflows and preventing unnecessary re-renders.
- Consolidated Redux fetch flows into centralized selectors to reduce redundant API traffic by ~30%, improving application throughput under high user load.
- Established Jest/RTL testing standards for core user flows (authentication, form validation), raising test coverage above 80% and catching critical bugs before production.
- Contributed to modernization and refactoring initiatives by enforcing Typescript standards, improving code quality and strengthening application stability.
- Designed AI-assisted workflow with python, Langchain and LangGraph for information retrieval, report generation and workflow automation use cases.
- Built RAG pipelines involving document ingestion, chunking, embeddings and vector-based retrieval along with semantic search, hybrid retrieval and reranking techniques to improve information retrieval effectiveness.
- Evaluated retrieval performance and experimented with response grounding, guardrails and prompt engineering techniques for AI applications.

PROJECTS

Multi-Agent AI Travel Planning System | Python, LangGraph, LangChain, Groq, PostgreSQL

- Multi-Agent AI Travel Planning System | Python, LangGraph, LangChain, Groq, PostgreSQL
- Developed a Multi-Agent AI application using LangGraph with specialized Flight, Hotel, Itinerary, and Final Response agents.
- Integrated Tavily Search API and AviationStack API to provide real-time web search and flight information.
- Designed a centralized shared state architecture for agent communication and workflow orchestration.
- Implemented persistent memory using PostgreSQL to maintain conversation history across sessions.
- Built modular agent pipelines using LangChain, improving maintainability and scalability.
- Applied prompt engineering and LLM orchestration with Groq Llama 3.3 for itinerary generation and response synthesis.

ACHIEVEMENTS & CERTIFICATIONS

- Microsoft Certified: Azure Fundamentals (AZ-900)**
- Microsoft Certified: Azure AI Fundamentals (AI-900)**
- EY AI Engineering – Bronze**
- Awarded “Going Above & Beyond” recognition at E&Y**

EDUCATION

D.Y. Patil College of Engineering (SPPU Affiliated)

Bachelor of Engineering - Computer Engineering (CGPA: 9.65) Pune, Maharashtra

Aug 2019 – July 2023